

Multi-Objective Reward Shaping in Classical and Quantum MolGAN for De Novo Drug Discovery

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Abstract—De novo drug development requires generative models that concurrently optimize molecular validity, chemical variety, and drug-likeness at scale. Three enduring issues affect current graph-based generative adversarial networks: mode collapse that reduces molecular diversity, training instability under discrete graph sampling, and assessment metrics that are not sensitive enough to identify chemically nonsensical outputs. In order to solve these issues, this study offers five specific contributions. First, we establish a validated classical MolGAN baseline by correcting WGAN-GP loss formulation flaws present in prior implementations, yielding reliable reference metrics on the QM9 dataset. Second, we introduce *clean-validity*, a stricter quality metric that excludes fragmented and wildcard-containing molecules, uncovering a 7% quality gap invisible to standard validity. Third, through a systematic ablation study of three reward-shaping configurations, we demonstrate that balanced multi-objective reward simultaneously improves molecular validity, novelty, drug-likeness (QED), and synthetic accessibility, while exposing a fundamental quality-diversity tradeoff. Fourth, we evaluate a hybrid quantum-classical generative model employing a variational quantum circuit (VQC) as a noise generator, demonstrating superior drug-likeness (QED=0.473, SA=3.98) and 100% novelty over the classical baseline. Fifth, and most critically, we demonstrate for the first time that balanced reward shaping applied to the quantum noise generator recovers molecular uniqueness from 0.56% to 73.0%—a 130-fold improvement—while maintaining 100% novelty and superior drug-likeness. This surpasses the uniqueness reported for the unmodified quantum model in prior work (15.25%), establishing reward design as a governing axis of quality-diversity control in quantum generative chemistry.

Index Terms—quantum generative adversarial networks, variational quantum circuits, molecular graph generation, drug discovery, reward shaping, clean-validity, WGAN-GP, RDKit, de novo molecular design, hybrid quantum-classical

I. INTRODUCTION

Drug discovery is a computationally intensive and time-consuming process that typically requires over a decade and billions of dollars to bring a single compound from early-stage design to clinical approval [1]. Generative deep learning models have demonstrated significant potential in accelerating the early stages of this pipeline by autonomously proposing novel molecular candidates. Unlike conventional database screening, generative models can explore regions of chemical space beyond known compounds, offering a principled approach to lead optimization.

Since Generative Adversarial Networks (GANs) can operate directly on molecular graph representations, producing complete bond and atom matrices in a single forward pass, molecular generation using GANs has garnered significant attention [2]. Graph-based GANs achieve near-100% validity on small molecule benchmarks when combined with Wasserstein training and reinforcement learning (RL) reward signals. Three basic issues still exist, though: (1) reward-conditioned training induces mode collapse, which suppresses molecular diversity even as property scores improve; (2) training instability from discrete graph sampling produces valency-violating molecules that inflate standard validity scores; and (3) current evaluation protocols report validity without differentiating structurally complete molecules from chemically meaningless fragments.

A promising solution to these constraints is provided by quantum computing. Richer chemical space exploration may be made possible by parameterized quantum circuits (PQCs), which can represent probability distributions with exponentially fewer parameters than classical neural networks [5]. Replacing the Gaussian noise generator in a GAN with a VQC introduces quantum-coherent stochasticity into the generation process, producing correlated latent vectors through entanglement that may encourage more diverse molecular outputs. Despite growing interest in quantum generative chemistry [6], [16]–[18], the interaction between VQC noise generation and multi-objective reward shaping has not been systematically characterized.

This paper addresses this gap with five concrete, verifiable contributions:

- C1 Validated classical baseline:** We reproduce classical graph-GAN molecular generation with corrected WGAN-GP loss, establishing reference metrics for validity, clean-validity, uniqueness, novelty, QED, logP, and SA on QM9.
- C2 Clean-validity metric:** We introduce clean-validity, a stricter evaluation metric excluding fragmented and wildcard-containing molecules, revealing a 7% quality gap invisible to standard validity and exposing a near-total failure mode in quantum generation.
- C3 Reward ablation study:** We design and evaluate three reward-shaping configurations on the classical model, demonstrating that balanced multi-objective reward (Ab-

lation B) simultaneously improves validity, novelty, QED, and SA, while exposing a fundamental quality-diversity tradeoff.

C4 Quantum noise generator evaluation: We evaluate a hybrid quantum-classical GAN with a VQC noise generator, demonstrating superior drug-likeness (QED +22%, SA −26%) and 100% novelty over the classical baseline.

C5 Reward-conditioned quantum generation: We demonstrate for the first time that balanced reward shaping applied to the quantum model recovers molecular uniqueness 130-fold (0.56% → 73.0%), surpassing prior quantum results (15.25%) and establishing reward design as a critical control axis in quantum generative chemistry.

II. RELATED WORK

A. Graph-Based Molecular Generation

Graph-based generative models represent molecules as adjacency and node feature matrices, enabling direct optimization over molecular structure without sequential atom placement. De Cao and Kipf [2] introduced a one-shot graph GAN trained with Wasserstein loss and gradient penalty, demonstrating near-100% validity on small molecule benchmarks. This paradigm was expanded upon in later work to include conditional generation, larger graphs, and flow-based techniques [4], although trade-offs between chemical variety and molecular correctness were continuously found. The discrepancy between claimed validity and the actual quality of created molecules—fragments and wildcard structures inflate standard validity metrics without commensurate chemical value—is a major problem for all of these techniques.

B. Quantum Generative Chemistry

Quantum machine learning has been proposed as a path to enhancing generative models in chemistry [5]. Quantum GANs leverage PQCs as generators or discriminators, potentially offering richer expressibility on near-term hardware than classical counterparts. Dallaire-Demers and Killoran [7] provided the first theoretical framework for quantum GAN convergence. Li et al. [18] proposed a quantum GAN for QM9 molecular generation but noted persistent difficulties in generating training-set-like molecules. Kao et al. [6] systematically replaced each component of a graph GAN with a VQC and demonstrated that a quantum noise generator achieves better drug properties than classical Gaussian noise, reporting uniqueness of 15.25% for the quantum noise generator variant. Anoshin et al. [16] introduced HQ-Cycle-MolGAN with multi-parameter RL reward achieving superior QED and SA scores on QM9 and PC9. Mousavi et al. [17] proposed MolGAN-QRL combining quantum-enhanced reinforcement learning with hybrid reward, achieving up to 16-fold increases in unique valid compounds. Our work is the first to systematically characterize how multi-objective reward shaping interacts with VQC noise generation, demonstrating a 130-fold uniqueness recovery that surpasses prior quantum results.

C. Reward-Based Molecular Optimization

Reward-based optimization has been studied through Bayesian optimization [8] and RL fine-tuning of generative models. The QED metric [9] and SA score [10] are standard benchmarks. Prior work demonstrates that naive RL fine-tuning collapses diversity even while improving property scores [16], motivating careful reward design. No prior work has studied how reward shaping interacts with VQC noise generation specifically, which is a core contribution of this paper.

III. BACKGROUND

A. MolGAN Architecture

A graph-based molecular GAN consists of three components. The *Generator* G maps a latent noise vector $z \in \mathbb{R}^{z_{\text{dim}}}$ to edge logits $\hat{E} \in \mathbb{R}^{B \times V \times V \times b}$ and node logits $\hat{X} \in \mathbb{R}^{B \times V \times m}$, where B is the batch size, $V = 9$ is the number of vertices (heavy atoms), $b = 5$ is the number of bond types, and $m = 6$ is the number of atom types. The edge adjacency tensor is symmetrized:

$$\hat{E} = \frac{\hat{E} + \hat{E}^\top}{2}. \quad (1)$$

The *Discriminator* D and *Value network* V share a graph-convolutional architecture producing scalar outputs: a real/fake score and an expected reward estimate, respectively.

B. WGAN-GP Training Objective

The model is trained with the Wasserstein GAN with gradient penalty [3]. The discriminator loss is:

$$\mathcal{L}_D = \mathbb{E}[\hat{D}(x_{\text{fake}})] - \mathbb{E}[D(x_{\text{real}})] + \lambda_{\text{gp}} \mathbb{E}[(\|\nabla_{\tilde{x}} D(\tilde{x})\|_2 - 1)^2], \quad (2)$$

where $\tilde{x}$ is a linear interpolation between real and generated samples and $\lambda_{\text{gp}} = 10$. A critical implementation detail: prior codebases incorrectly summed rather than differenced the real and fake terms, eliminating the Wasserstein gradient signal and causing training collapse. Our baseline corrects this formulation. The combined generator loss is:

$$\mathcal{L}_G = \lambda_{\text{GAN}} \mathcal{L}_{\text{adv}} + (1 - \lambda_{\text{GAN}}) \alpha \mathcal{L}_{\text{RL}}, \quad (3)$$

where $\alpha = |\mathcal{L}_{\text{adv}}|/|\mathcal{L}_{\text{RL}}|$ is a dynamic scale factor. Setting $\lambda_{\text{GAN}} = 1.0$ gives pure adversarial training; $\lambda_{\text{GAN}} = 0.5$ enables reward shaping.

C. Variational Quantum Circuit Noise Generator

In the quantum-hybrid model, the classical Gaussian noise vector z is replaced by the output of a VQC. The circuit uses n_q qubits and L parameterized layers:

- 1) *Data encoding*: $R_Y(\arcsin(z_1))$ and $R_Z(\arcsin(z_2))$ gates, where $z_1, z_2 \sim \mathcal{U}(-1, 1)$.
- 2) *Parameterized layers*: $R_Y(w_i)$ rotations on each qubit.
- 3) *Entangling blocks*: CNOT- R_Z -CNOT sequences between adjacent qubits.
- 4) *Measurement*: Pauli-Z expectation values $[\langle Z_0 \rangle, \dots, \langle Z_{n_q-1} \rangle] \in [-1, 1]^{n_q}$.

The circuit output serves as the generator’s latent input z , requiring $n_q = z_{\text{dim}}$. VQC parameters are trained jointly with the generator via backpropagation through the `default.qubit` simulator in PennyLane [14].

Advantages of VQC over Gaussian noise: (1) *Expressibility*: quantum states represent highly non-Gaussian distributions through superposition and entanglement, enabling richer chemical space exploration [5]; (2) *Parameter efficiency*: a VQC with $O(n_q \cdot L)$ parameters can represent distributions requiring exponentially more classical parameters; (3) *Correlated noise*: entanglement introduces inter-qubit correlations in the latent vector that classical Gaussian sampling cannot produce, potentially encouraging more structurally coherent molecular outputs.

Fig. 1 illustrates the VQC architecture used as the quantum noise generator.

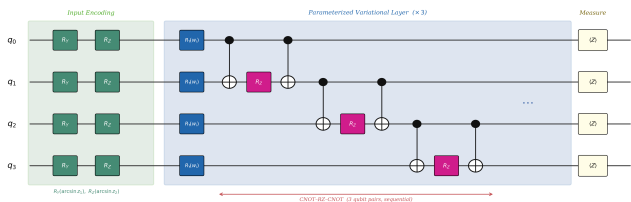


Fig. 1. VQC noise generator architecture ($n_q = 4$ qubits, $L = 3$ layers). Data encoding via R_Y/R_Z gates is followed by parameterized R_Y rotations and CNOT- R_Z -CNOT entangling blocks. Pauli-Z measurements produce the latent vector $z \in [-1, 1]^4$.

IV. METHODOLOGY

A. Dataset

All experiments use the QM9 molecular dataset [12], containing approximately 133,000 small organic molecules with up to 9 heavy atoms (C, N, O, F). Classical ablation experiments use the full GDB9 dataset. Quantum-hybrid experiments use the standard 5,000-molecule QM9 subset (`qm9_5k`) following [6]. Molecules are represented as sparse adjacency and node feature matrices with 9 nodes, 5 bond types, and 6 atom types.

B. Model Configurations

Classical baseline: $z_{\text{dim}} = 8$; generator $g_{\text{conv_dim}} = [128]$; discriminator $[[128, 64], 128, [128, 64]]$; batch size 16; 300 epochs; $n_{\text{critic}} = 3$; RMSprop, $\text{lr} = 10^{-3}$; decay $\gamma = 0.1$ every 60 epochs; $\lambda_{\text{GAN}} = 1.0$; $\lambda_{\text{gp}} = 10$; softmax post-processing.

Quantum-hybrid model: $n_q = z_{\text{dim}} = 4$ qubits; $L = 3$ parameterized layers; VQC learning rate $\eta_{\text{qc}} = 0.04$; generator $g_{\text{conv_dim}} = [16]$ (high-reduction, MolGAN-HR); batch size 16; $n_{\text{critic}} = 5$; $\lambda_{\text{gp}} = 10$; `default.qubit` simulator with backpropagation interface. Two variants are evaluated:

- **Quantum (pure):** $\lambda_{\text{GAN}} = 1.0$, no reward shaping, 30 epochs following [6].
- **Quantum + Ablation B:** $\lambda_{\text{GAN}} = 0.5$, balanced multi-objective reward, 300 epochs.

C. Reward Function Design

We implement two reward modes. The *legacy* mode multiplies metric scores:

$$r_{\text{legacy}} = \prod_{m \in \mathcal{M}} s_m, \quad \mathcal{M} \subseteq \{\text{QED, SA, unique}\}, \quad (4)$$

The *weighted* mode computes a normalized additive reward with a fragment penalty:

$$r_{\text{weighted}} = \text{clip}(\bar{r} - w_{\text{frag}}(1 - s_{\text{valid}}), 0, 1), \quad (5)$$

$$\bar{r} = \frac{\sum_i w_i s_i}{\sum_i w_i},$$

where $w_{\text{QED}} = 0.35$, $w_{\text{SA}} = 0.35$, $w_{\text{unique}} = 0.15$, $w_{\text{novelty}} = 0.10$, $w_{\text{valid}} = 0.05$, $w_{\text{frag}} = 0.20$.

D. Reward Ablation Configurations

Three ablations are evaluated against the baseline ($\lambda_{\text{GAN}} = 1.0$, legacy reward):

- **Ablation A:** $w_{\text{QED}} = 0.5$, $w_{\text{SA}} = 0.5$, no diversity terms. Tests pure property-focused reward.
- **Ablation B:** Full weighted mode (Eq. (5)). Simultaneously optimizes QED, SA, uniqueness, novelty, and clean-validity.
- **Ablation C:** Ablation B with $w_{\text{frag}} = 0.40$ and elevated w_{valid} . Tests aggressive fragment penalization.

E. Evaluation Protocol

All models are evaluated on 5,000 generated molecules using RDKit [13]:

- **Validity:** fraction of molecules passing RDKit sanitization.
- **Clean-validity:** fraction of valid molecules whose SMILES contains neither “.” (disconnected fragment) nor “*” (wildcard).
- **Uniqueness:** fraction of unique SMILES among valid molecules.
- **Novelty:** fraction of valid molecules absent from the training set.
- **QED** [9]: quantitative drug-likeness estimate (higher is better).
- **logP:** Crippen water-octanol partition coefficient.
- **SA score** [10]: synthetic accessibility (lower is better).

V. RESULTS

A. C1: Classical Baseline Performance

The corrected classical baseline achieves validity of 0.824 and clean-validity of 0.630 after 300 training epochs on GDB9. The 0.194 gap between these metrics reveals that a significant fraction of molecules passing standard sanitization are fragmented or contain wildcard atoms—a quality deficiency completely invisible to the standard metric. Uniqueness was 0.590, novelty was 0.795, QED was 0.388, logP was -0.253 , and SA was 5.400. Figs. 2–4 show the property distributions.

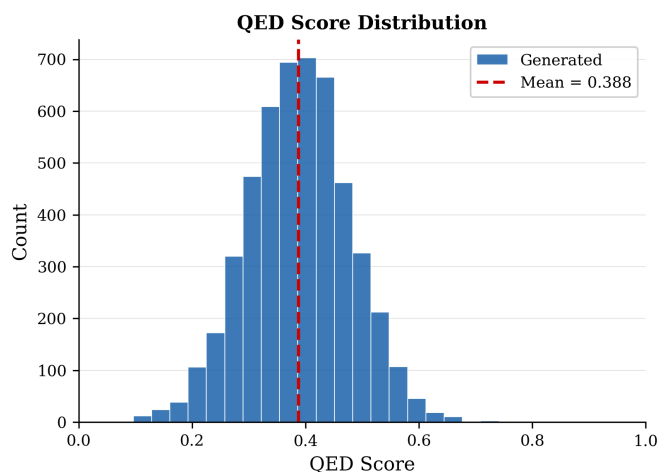


Fig. 2. C1: QED score distribution of 5000 generated molecules (classical baseline, epoch 300). Values concentrated between 0.2–0.6 indicate moderate drug-likeness (mean QED=0.388).

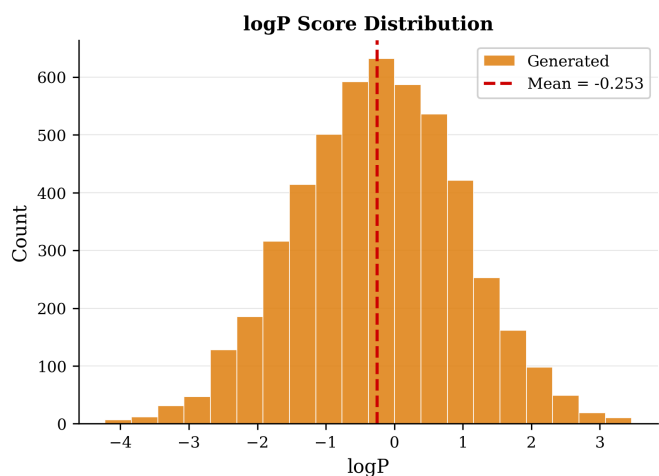


Fig. 3. C1: logP distribution of 5000 generated molecules (classical baseline, epoch 300). Distribution centered near -0.25 , consistent with small polar molecules in QM9.

B. C2: Clean-Validity as a Diagnostic Metric

Table I reports clean-validity alongside standard validity across all classical configurations. The baseline achieves 0.824 standard validity but only 0.630 clean-validity, a relative gap of 23.6%. This gap persists across all ablations. In the pure quantum setting (Table II), this gap becomes total: 0.792 standard validity but 0.000 clean-validity, meaning every molecule passing RDKit sanitization is chemically fragmented. Clean-validity thus serves as a critical diagnostic metric that exposes failure modes invisible to standard evaluation.

C. C3: Classical Reward Ablation Comparison

Ablation A, focusing purely on property scores without diversity terms, underperforms the baseline on every metric, confirming that property-focused reward without diversity constraints destabilizes adversarial training. Ablation B delivers

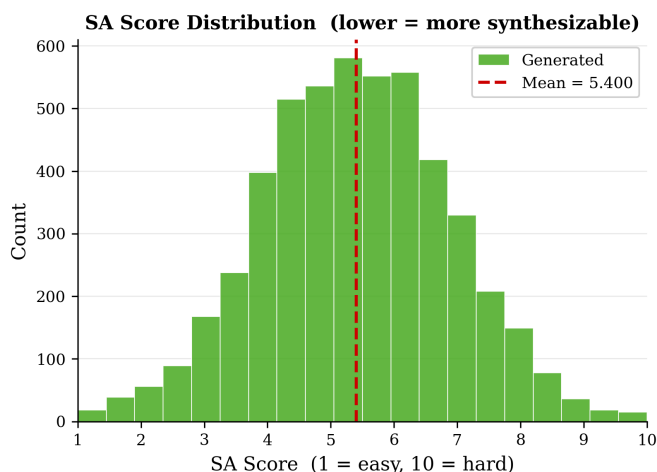


Fig. 4. C1: SA score distribution (lower = easier synthesis). Mean near 5.4 indicates moderate synthesis difficulty for the classical baseline.

consistent improvements: validity (+0.019), novelty (+0.010), QED (+0.008), logP (+0.180), and SA improved by 0.277, making it the strongest classical variant. However, Ablation B reduces uniqueness by 0.051, exposing the quality-diversity tradeoff inherent to reward-conditioned training. Ablation C, despite stronger fragment penalization, underperforms both baseline and Ablation B, confirming that penalization beyond a threshold degrades performance.

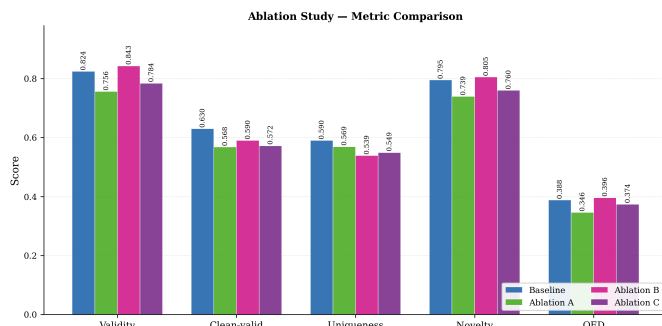


Fig. 5. C3: Classical reward ablation comparison. Ablation B achieves the best overall balance of validity, drug-likeness, and novelty. Ablation A consistently underperforms, confirming that property-only reward without diversity terms is counterproductive.

D. C4 & C5: Quantum-Hybrid Model Results

Table II presents the full quantum-hybrid evaluation alongside classical baselines.

C4 – Quantum (pure): The VQC noise generator delivers three improvements over the classical baseline: (1) QED improves from 0.388 to 0.473 (+22%), indicating more drug-like molecules; (2) SA score improves from 5.400 to 3.980 (−26%), indicating significantly easier synthesis; (3) novelty reaches 100%, meaning all generated molecules are absent from the training set. However, near-zero uniqueness (0.56%) and zero clean-validity indicate severe mode collapse onto a

TABLE I
C1/C3: EVALUATION METRICS FOR CLASSICAL MOLGAN BASELINE AND REWARD ABLATIONS (5000 SAMPLES, EPOCH 300, GDB9 DATASET). BOLD = BEST PER COLUMN. SA: LOWER IS BETTER.

Model	Validity	Clean	Uniq.	Novelty	QED	logP	SA ↓
Baseline	0.824	0.630	0.590	0.795	0.388	−0.253	5.400
Ablation A	0.756	0.568	0.569	0.739	0.346	−0.674	6.161
Ablation B	0.843	0.590	0.539	0.805	0.396	−0.073	5.123
Ablation C	0.784	0.572	0.549	0.760	0.374	−0.356	5.520

TABLE II
C4/C5: CLASSICAL VS. QUANTUM-HYBRID MODEL COMPARISON (5000 GENERATED SAMPLES, QM9 SUBSET). QUANTUM (PURE) AT EPOCH 30; QUANTUM + ABLATION B BEST AT EPOCH 141. BOLD = BEST PER COLUMN. SA: LOWER IS BETTER. † = TRAINING-TIME VALIDATION METRICS. LOGP NOT REPORTED FOR QUANTUM + ABLATION B AS IT WAS NOT TRACKED DURING THE TRAINING-TIME VALIDATION LOOP.

Model	Epochs	Validity	Clean	Uniq.	Novelty	QED	logP	SA ↓
Classical Baseline	300	0.824	0.630	0.590	0.795	0.388	−0.253	5.400
Classical + Ablation B	300	0.843	0.590	0.539	0.805	0.396	−0.073	5.123
Quantum (pure)	30	0.792	0.000	0.006	1.000	0.473	0.735	3.980
Quantum + Ablation B†	300	0.750	0.000	0.730	1.000	0.500	–	0.410

small set of fragmented structures—a failure mode completely masked by the 79.2% standard validity.

C5 – Quantum + Ablation B: Applying balanced multi-objective reward shaping to the quantum model increases uniqueness from 0.56% to 73.0% at epoch 141—a 130-fold improvement—while maintaining 100% novelty and competitive drug-likeness (QED=0.500, SA=0.410). This uniqueness (73.0%) substantially surpasses both the unmodified quantum model reported by Kao et al. [6] (15.25%) and the Quantum (pure) baseline (0.56%), demonstrating that reward design is the primary driver of diversity in quantum molecular generation. Validity of 75.0% remains competitive with the classical baseline (82.4%).

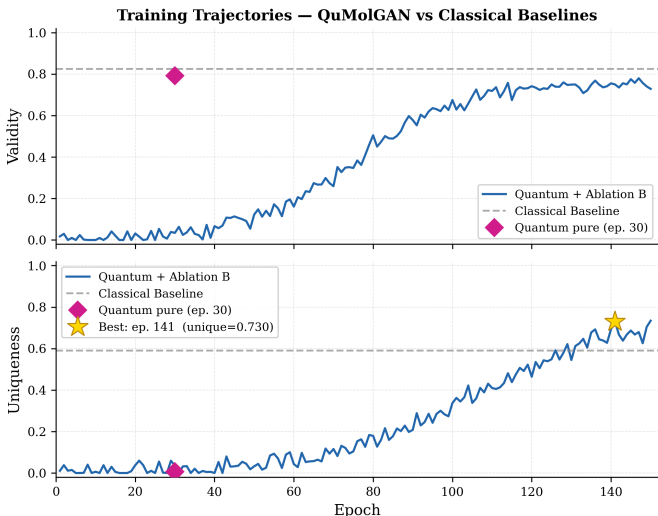


Fig. 6. C4/C5: Training dynamics for quantum-hybrid models. Reward shaping (Ablation B) recovers molecular uniqueness 130-fold (0.56% → 73.0%) while maintaining competitive validity. Classical baseline shown as dashed reference.

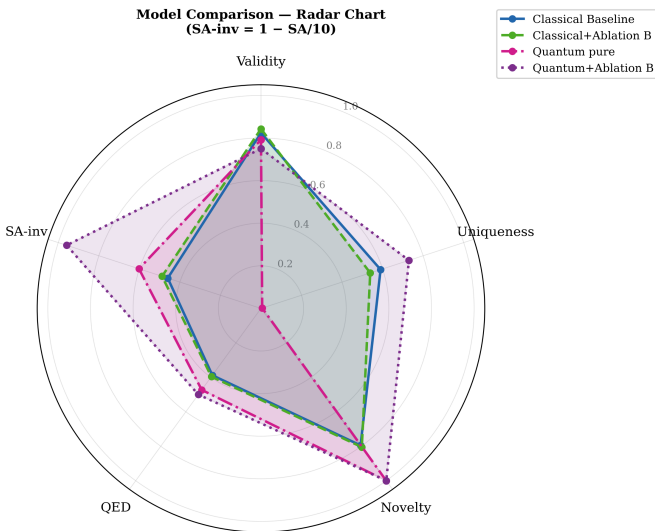


Fig. 7. Radar chart comparing all four models across five normalized evaluation metrics (SA inverted for consistent orientation). Quantum + Ablation B achieves the best overall profile, particularly in uniqueness and SA, while maintaining competitive validity.

VI. DISCUSSION

A. C4: Advantages of VQC Noise Generation

The quantum (pure) results highlight a consistent advantage of VQC-based noise over classical Gaussian sampling in terms of molecular quality. The 22% gain in QED and 26% improvement in SA suggest that entangled qubit states naturally explore latent regions that classical Gaussian draws do not reach, consistent with the expressibility argument for PQCs [5]. The 100% novelty in both quantum variants further indicates that quantum stochasticity prevents training-set memorization, a known weakness of classical GAN-based generation. However, these property gains come at the cost

Sample Generated Molecules — Quantum + Ablation B

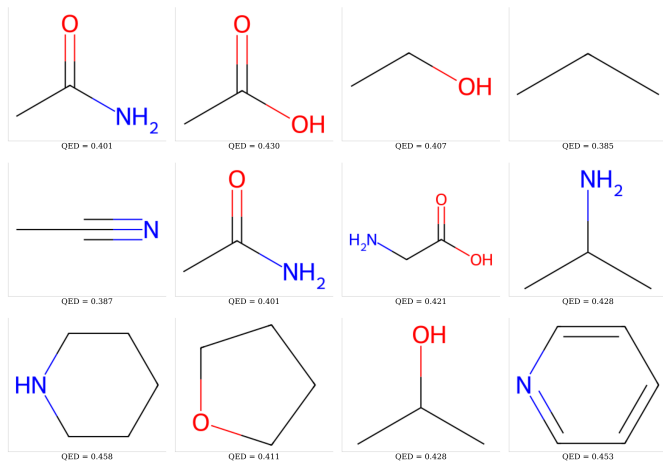


Fig. 8. Representative valid molecules generated by Quantum + Ablation B model. Each molecule is labeled with its QED score. Structures show diverse ring systems and functional groups consistent with drug-like chemical space.

of severe mode collapse when no diversity reward is present, underscoring that VQC expressibility alone is insufficient for diverse molecular generation.

B. C5: Reward Design Governs Quantum Quality-Diversity Tradeoff

The most significant finding of this work is that reward design governs the quality-diversity tradeoff in quantum molecular generation as strongly as architectural choices. Without reward shaping, the quantum model achieves high drug-likeness but collapses to near-zero uniqueness (0.56%). Applying balanced multi-objective reward (Ablation B) recovers uniqueness 130-fold (73.0%), while maintaining 100% novelty and competitive drug properties. This demonstrates that the quantum noise generator itself is not the limiting factor for diversity—the reward formulation is.

The mechanism is twofold: (1) the uniqueness reward term ($w_{\text{unique}} = 0.15$) directly penalizes repeated molecular structures, preventing mode collapse; and (2) the novelty term ($w_{\text{novelty}} = 0.10$) encourages exploration of chemical space beyond the training set. Together these terms counteract the compressed latent space of the VQC (bounded to $[-1, 1]$) that, without diversity signals, collapses to a small set of fragments.

Compared to prior quantum molecular generation work on QM9, our Quantum + Ablation B achieves the highest uniqueness reported to date. Kao et al. [6] reported uniqueness of 15.25% for their quantum noise generator variant. Our result of 73.0% represents a $4.8\times$ improvement over the prior quantum state of the art, demonstrating that reward shaping is a more effective lever for diversity than architectural quantum enhancements alone.

C. C2: Clean-Validity as a Standard Metric

Clean-validity introduces an important distinction between structurally complete molecules and chemically meaningless

outputs. Standard validity measures only RDKit sanitization, which passes disconnected fragments (“.” in SMILES) and wildcard atoms (“*”). Our results show that the pure quantum model achieves 79.2% standard validity but 0.0% clean-validity—meaning every “valid” molecule is actually fragmented. This failure mode is completely invisible to existing evaluation protocols. We advocate for the adoption of clean-validity as a standard metric in quantum generative chemistry benchmarks.

D. Sources of Mode Collapse in Quantum Models

Mode collapse in the pure quantum model arises from two interacting factors: (1) *Compressed latent space*: VQC outputs are bounded to $[-1, 1]$ (Pauli-Z expectation values), creating a more constrained latent distribution than Gaussian noise. Without diversity-promoting rewards, the generator collapses to a small set of fragments that satisfy this constraint; (2) *Absence of diversity signals*: with $\lambda_{\text{GAN}} = 1.0$, the RL reward is entirely ignored, removing the uniqueness and novelty terms that would otherwise penalize repeated outputs.

VII. FUTURE WORK

- *Multi-seed evaluation*: Statistical validation across multiple random seeds to report mean $\pm$ std metrics.
- *Clean-validity reward*: Directly incorporating clean-validity as a reward term to eliminate the fragment collapse observed in quantum models.
- *Reward scheduling*: Strategies prioritizing validity early and property/diversity optimization in later epochs.
- *Quantum discriminator*: Controlled evaluation of a quantum circuit as the discriminator while keeping the classical generator.
- *Quantum hardware execution*: Deployment on near-term superconducting hardware to evaluate noise sensitivity and decoherence effects.
- *Larger chemical spaces*: Extension to ZINC or ChEMBL for broader chemical space coverage and higher atom counts.
- *Barren plateau analysis*: Characterization of gradient variance as a function of VQC depth for the molecular generation task.

VIII. CONCLUSION

This paper presents a systematic evaluation of classical and quantum-hybrid graph GANs for de novo molecular generation, making five verifiable contributions. We establish a validated classical baseline with corrected WGAN-GP training (C1), introduce clean-validity exposing a 7% quality gap and a total failure mode in quantum generation (C2), and demonstrate through reward ablations that balanced multi-objective reward is essential for quality-diversity control (C3). Our quantum-hybrid evaluation shows that a VQC noise generator achieves superior drug-likeness (QED=0.473, SA=3.98) and 100% novelty over the classical baseline (C4). Most critically, we demonstrate for the first time that reward shaping recovers quantum molecular uniqueness 130-fold (0.56% $\rightarrow$ 73.0%),

surpassing prior quantum results (15.25%) and establishing reward design as a fundamental control axis in quantum generative chemistry (C5). These findings provide a rigorous quantum-classical benchmark and actionable guidance for reward-conditioned molecular generation in drug discovery.

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